Alien Perspective on Analysis

The blog was inspired by the following thought process:

> *We start with the natural numbers, construct the integers, then the rationals. We complete them to get the reals, and finally adjoin i to get the complex numbers. This is the "standard" way. But is it the* only *way?*

Imagine an alien civilization that understands number theory perfectly but has no physical intuition for "continuity" or "measuring lengths." To them, the algebraic closure of the rationals, $\overline{\mathbb{Q}}$, is the natural universe. This blog explores how such a civilization would discover the real numbers not as a foundation, but as a strange, derived sub-structure, and how this would change their understanding of calculus.

Be warned that this blog is written with "free-flow" writing and not too much thought was put in to the structure. I will not really prove anything and not define all my terms; I will just assume proficiency with most of the background material. All the theorems and relevant definitions can be found in either [4], [1], [2], [5], or [3].

1 Constructing Reality from Algebra

Our aliens start with $\mathbb{Q}$. Their natural next step is to solve all polynomial equations, leading them to $\overline{\mathbb{Q}}$. They have no topology yet. How do they find $\mathbb{R}$? Classically, we view $\mathbb{C}$ as a degree 2 extension of $\mathbb{R}$. The aliens view this in reverse: they search for a subfield of index 2 inside their algebraic closure.

Theorem 1.1: Artin-Schreier

Let K be an algebraically closed field. If $F \subset K$ is a subfield of finite index (i.e., $1 < [K : F] < \infty$), then:

1. $[K : F] = 2$.
2. $K = F(i)$ where $i^2 = -1$.
3. F is a *Real Closed Field*.

To find "Reality," the aliens look at the symmetries of their universe: the Absolute Galois Group $G = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. They seek an *involution* $\sigma \in G$ (an automorphism where $\sigma^2 = \mathrm{id}$).

Definition 1.2: The Alien Construction of $\mathbb{R}$

Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be an involution. We define the "Real Algebraic Numbers" relative to σ as the fixed field:

$$F_\sigma = \{x \in \overline{\mathbb{Q}} \big| \sigma(x) = x\}$$

> **Note**: This construction is *not unique*. There are infinitely many involutions in the Galois group, all conjugate to each other. The alien can build *a* real line, but they cannot distinguish *which* one is "ours" without making an arbitrary choice. To them, $\sqrt[3]{2}$ might be "real" in one coordinate system, but "complex" in another.

1.1 The Topological Completion

Once F_σ is chosen, the ordering is algebraic: $x \geq 0 \iff \exists y \in F_\sigma, x = y^2$. But how do they recover $\mathbb{C}$? They cannot use a topology intrinsic to $\overline{\mathbb{Q}}$ alone. Instead, they define a *norm* relative to their chosen involution.

Since F_σ is real closed, the sum of two squares is always a square. The alien defines the "σ-norm" $|\cdot|_\sigma : \overline{\mathbb{Q}} \to F_\sigma$ as:

$$|z|_\sigma = \sqrt{z \cdot \sigma(z)}$$

Algebraically, if $z = x + iy$, then $z \cdot \sigma(z) = x^2 + y^2$, which has a square root in F_σ. This allows them to define a metric $d(a, b) = |a - b|_\sigma$.

They now run the standard completion process (Cauchy sequences):

1. Completing F_σ yields $\mathbb{R}$.
2. Completing $\overline{\mathbb{Q}}$ (which is a 2D vector space over F_σ) yields the product topology:

$$\widehat{\overline{\mathbb{Q}}} \cong \widehat{F_\sigma} \oplus \widehat{F_\sigma} i \cong \mathbb{R} \oplus \mathbb{R}i \cong \mathbb{C}$$

2 The Archimedean Anomaly

The topology derived above—the Order Topology—would be viewed by the aliens as algebraically natural, but geometrically "dangerous."

2.1 The Topology of Squares

The aliens admit the definition is elegant because it requires no new concepts. They don't need a visualization of a number line; they just use squares. Rather than defining open intervals (a, b) via inequalities, they define the basis of the topology as:

$$U_{a,b} = \{x \in F \big| (x - a) \in (F^\times)^2 \text{ and } (b - x) \in (F^\times)^2\}$$

Because F is a Real Closed Field, the "positive" elements are exactly the squares. So, the alien sees the order topology not as "left vs. right," but as a statement about the solvability of quadratic equations. To be "in the interval" means you can take the square root of your distance to the endpoints.

2.2 The Singularity

However, once they start studying this topology, they find it completely bizarre compared to the rest of their universe ($\mathbb{Q}_p$, finite fields).

Most valuations in algebra are **Non-Archimedean** (satisfying the strong triangle inequality $|x+y| \leq \max(|x|, |y|)$). This creates a stable, bounded universe. The Order Topology on F is the only one that is **Archimedean** ($|x+y| \leq |x| + |y|$).

- **Explosive Growth**: F is the only place where adding small things together eventually creates an arbitrarily large monster. In $\mathbb{Q}_p$, "infinity" is not a direction you can reach by walking. In F, it is.
- **Directedness**: The existence of the order creates an asymmetry. In $\mathbb{C}$ or $\mathbb{Q}_p$, x and $-x$ are topologically indistinguishable. In F, there is a fundamental difference between "positive" (a square) and "negative" (not a square).

To the alien, the universe is mostly disconnected dust ($\mathbb{Q}_p$), but there is one **singular point at infinity** ($\mathbb{R}$) where the dust fuses together, explodes in size, and gains a direction.

3 The Rigidity Paradox

The standard modern intuition is to consider real calculus (smooth functions, bump functions) to be the standard, and complex calculus (holomorphic functions) to be a restrictive special case satisfying the Cauchy-Riemann equations.

To the alien, this is backward. The "Natural" calculus is on $\overline{\mathbb{Q}}$ (or $\mathbb{C}$): a function $f : \mathbb{C} \to \mathbb{C}$ is differentiable if it is locally linear *over the field* $\mathbb{C}$. This means the local approximation must be multiplication by a scalar $\lambda \in \mathbb{C}$.

If we translate this requirement into the language of the fixed field $\mathbb{R} \subset \mathbb{C}$, the requirement that the Jacobian matrix acts like a complex number *is* the Cauchy-Riemann equations.

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \iff \text{Cauchy-Riemann}$$

There is another reason the alien considers Cauchy-Riemann functions the only "natural" ones. Remember, they constructed their coordinates by making an *arbitrary choice* of σ.

A "natural" function (like a polynomial $P(z) = z^2 + 1$) is defined using only field operations; it does not care which σ you chose. A "non-natural" function (like $f(z) = \mathrm{Re}(z) = \frac{z+\sigma(z)}{2}$) requires you to know exactly which involution was chosen.

The Cauchy-Riemann equations can be written in Wirtinger derivative notation as $\frac{\partial f}{\partial \bar{z}} = 0$. Translated to Alien-speak:

$$\frac{\partial f}{\partial z^\sigma} = 0$$

To the alien, the Cauchy-Riemann equations are the statement: *"This function depends only on the number z itself, and not on the arbitrary choice of involution used to construct the coordinate system."*

3.1 Why is Real Calculus "Freer"?

If $\mathbb{C}$ is the natural world, why does $\mathbb{R}$ allow for "wild" objects like bump functions (smooth functions that are non-zero in a small set and zero everywhere else)?

- In $\mathbb{C}$: To be smooth, you must satisfy the Cauchy-Riemann equations (Coordinate Independence). This "cross-check" enforces rigidity: if you know a function on a tiny open set, you know it everywhere (Analytic Continuation).
- In $\mathbb{R}$: By restricting to the fixed field, we effectively *turn off* the sensor in the imaginary dimension. We no longer require the derivative to match an approach from the i-direction.

The aliens view Smooth Real functions not as a richer class, but as a *broken* class. They are the pathological consequence of taking a slice of the universe and ignoring the ambient constraints of the algebraic closure.

4 Algebraic Differentiation: The Dual Numbers

Can the aliens define differentiation without limits or topology? Yes. They use the *Dual Numbers*.

Definition 4.1: Algebraic Derivative

Let R be a ring. Consider the ring of dual numbers $R[\varepsilon]/(\varepsilon^2)$. For a polynomial $P(x)$, we calculate:

$$P(x+\varepsilon) = P(x) + P'(x)\varepsilon$$

The coefficient of ε is the derivative.

This definition is purely algebraic. Does it descend to the Real subfield?

$$\begin{tikzcd} K[x] \arrow[r, "D"] \arrow[d, "\text{restrict}"'] & K[x] \arrow[d, "\text{restrict}"] \\ F[x] \arrow[r, "D"] & F[x] \end{tikzcd}$$

Since the derivative of a polynomial with coefficients in F has coefficients in F, the diagram commutes.

What is interesting about using automatic differentiation is that the input is forced to be a algebraic (polynomials, rational polynomials, series, etc.). The Algebraic Derivative is blind to the "smooth monsters" of real analysis. If you try to feed it a bump function (which has no Taylor series at the boundary), the algebraic machine jams. To the alien, bump functions are ghosts that appear only when you introduce topology.

5 The Spectrum of Topologies

Finally, let's take a look at Valuation Theory. They know that for every prime number p (and the infinite prime ∞), there is a valid, natural metric topology on $\overline{\mathbb{Q}}$. This gives them a whole spectrum of possible universes.

Which one is the "correct" one? They run a simple compatibility test against their constructed Real Field F_σ.

- **The p-adic Topology**: If they put a p-adic metric on $\overline{\mathbb{Q}}$, the subfield F_σ becomes totally disconnected dust. The algebraic order ($x > y$) has no relation to the topological closeness. The interval $[0, 1]$ is shredded.

- **The Archimedean Topology**: If they put the infinite-prime metric on $\overline{\mathbb{Q}}$, the subfield F_σ becomes a connected line. The topology respects the order.

The alien concludes that while $\overline{\mathbb{Q}}$ has many natural topologies, only the Archimedean one is compatible with the Order Structure of the fixed field.

Example 5.1: The Local Constancy Failure

Differentiation is a universal operator, but its meaning changes with the topology. In the p-adic worlds ($\mathbb{Q}_p$), the topology is totally disconnected. There exist locally constant functions that are not globally constant.

$$f'(x) = 0 \not\Rightarrow f(x) = \text{const}$$

In $\mathbb{Q}_\infty$ ($\mathbb{R}$), the Archimedean property forces the topology to be connected.

$$f'(x) = 0 \implies f(x) = \text{const}$$

The aliens conclude that $\mathbb{R}$ is not special just because of its order, but because it is the *Connected Field*. It is the only completion of $\mathbb{Q}$ where the "micro-behavior" (derivative) forces the "macro-behavior" (continuity) to align.

This connectivity ascends to the algebraic closure, $\mathbb{C}$. Because the topology of $\mathbb{C}$ is just the product topology of $\mathbb{R}$, $\mathbb{C}$ inherits this cohesion. This is the structural reason why *Analytic Continuation* works in the Archimedean world (a function defined on a small disk determines the function everywhere) but fails in the p-adic worlds. The "rigid"" structure of $\mathbb{C}$ is induced by the real order Topology.

References

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[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra.* 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis.* 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.